3.1 Introduction to Determinants

A 2×2 matrix is invertible iff its determinant is nonzero. To extend this useful fact to larger matrices, we need a definition for the determinant of an $n \times n$ matrix. We can discover the definition for the 3×3 case by watching what happens when an invertible 3×3 matrix A is row reduced.

Consider $A = [a_{ij}]$ with $a_{11} \neq 0$. If we multiply second and third rows of A by a_{11} and then subtract appropriate multiples of first row from the other two rows, we find out that:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \sim \begin{matrix} a_{11} \cdot R_2 \to \\ a_{11} R_3 \to \end{matrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{11} \cdot a_{21} & a_{11} \cdot a_{22} & a_{11} a_{23} \\ a_{11} \cdot a_{31} & a_{11} \cdot a_{32} & a_{11} \cdot a_{33} \end{bmatrix}$$

$$\sim \begin{matrix} R_2 - a_{21} \cdot R_1 \to \\ R_3 - a_{31} \cdot R_1 \to \end{matrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{11} \cdot a_{22} - a_{12} \cdot a_{21} & a_{11} a_{23} - a_{13} a_{21} \\ 0 & a_{11} \cdot a_{32} - a_{12} \cdot a_{31} & a_{11} \cdot a_{33} - a_{13} \cdot a_{31} \end{bmatrix}$$

Since A is invertible $(2,2)$ entry or the $(3,2)$ is nonzero. Suppose $(2,2)$ entry is nonzero. Multiply row 3 by $a_{11}a_{22} - a_{12}a_{21}$, and then to the new row 3 add $-(a_{11}a_{32} - a_{12}a_{31})$ times row 2. This will show that

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{11}a_{22} - a_{12}a_{21} & a_{11}a_{23} - a_{13}a_{21} \\ 0 & 0 & a_{11}\Delta \end{bmatrix}$$

where

$$\Delta = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}$$

Since A is invertible Δ must be nonzero. We call Δ as the determinant of the 3x3 matrix A.

- The determinant of a 2x2 matrix, A, is the number $\det A = a_{11}a_{22} - a_{12}a_{21}$.
- For a 1x1 matrix, A, we define $\det A = a_{11}$.
- To generalize the definition of the determinant to larger matrices, we'll use 2x2 determinants to rewrite the 3x3 determinant Δ described above. Since the terms in Δ can be grouped as

$$(a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32}) - (a_{12}a_{21}a_{33} - a_{12}a_{23}a_{31}) + (a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31})$$

$$\Delta = a_{11}\det\underbrace{\begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}}_{A_{11}} - a_{12}\det\underbrace{\begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}}_{A_{12}} + a_{13}\det\underbrace{\begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}}_{A_{13}}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Some definition to be used in Δ:

For any square matrix B, let B_{ij} denote the submatrix formed by deleting the ith row and jth column of B. For instance if

$$B = \begin{bmatrix} 1 & -2 & 5 & 0 \\ 2 & 0 & 4 & -1 \\ 3 & 1 & 0 & 7 \\ 0 & 4 & -2 & 0 \end{bmatrix} \quad \text{then} \quad B_{32} = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

Then, we can rewrite determinant of the A matrix as follows:

$$\Delta = a_{11} \cdot \det A_{11} - a_{12} \cdot \det A_{12} + a_{13} \det A_{13}$$

we can now give a recursive definition of a determinant. when $n=3$, $\det A$ is defined using determinants of the 2×2 submatrices A_{ij}. when $n=4$, $\det A$ use determinants of the 3×3 submatrices A_{ij}. In general an $n\times n$ determinant is defined by determinants of $(n-1)\times(n-1)$ submatrices.

Definition: For $n \geq 2$, the determinant of an $n \times n$ matrix $A = [a_{ij}]$ is the sum of n terms of the form $\mp a_{1j} \cdot \det A_{1j}$, with plus and minus signs alternating, where the entries $a_{11}, a_{12}, \ldots, a_{1n}$ are from the first row of A. In symbols,

$$\det A = a_{11} \det A_{11} - a_{12} \det A_{12} + \cdots + (-1)^{1+n} a_{1n} \det A_{1n}$$

$$= \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \cdot \det A_{1j}$$

Ex: Compute the determinant of $A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$

Sol:

$$\det A = (-1)^{1+1} \cdot 1 \cdot \det \begin{bmatrix} 4 & -1 \\ -2 & 0 \end{bmatrix} + (-1)^{1+2} \cdot 5 \cdot \det \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix} + (-1)^{1+3} \cdot 0 \cdot \det \begin{bmatrix} 2 & 4 \\ 0 & -2 \end{bmatrix}$$

$$= 1 \cdot (4 \cdot 0 - (-1)(-2)) + (-1) \cdot 5 \cdot (2 \cdot 0 - 0 \cdot -1) + 0$$

$$= 1 \cdot (0 - 2) - 5 (0 - 0) + 0 = -2$$

Another common notation is

$$\det A = 1 \cdot \begin{vmatrix} 4 & -1 \\ -2 & 0 \end{vmatrix} - 5 \cdot \begin{vmatrix} 2 & -1 \\ 0 & 0 \end{vmatrix} + 0 \cdot \begin{vmatrix} 2 & 4 \\ 0 & -2 \end{vmatrix}$$

Def: Given $A = [a_{ij}]$, the (i,j)-cofactor of A is the number C_{ij} given by $C_{ij} = (-1)^{i+j} \det A_{ij}$

Then $\det A = a_{11} C_{11} + a_{12} C_{12} + \cdots + a_{1n} C_{1n}$

This formula is called a cofactor expansion across the first row of A.

Theorem: The determinant of an $n \times n$ matrix A can be computed by a cofactor expansion across any row or down any column.

The expansion across the ith row using the cofactors is

$$\det A = a_{i1} C_{i1} + a_{i2} C_{i2} + \cdots + a_{in} C_{in}$$

The cofactor expansion down the jth column is:

$$\det A = a_{1j} C_{1j} + a_{2j} C_{2j} + \cdots + a_{nj} C_{nj}$$

The factor $(-1)^{i+j}$ determines the following checkerboard pattern of signs:

$$\begin{bmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Ex: Use a cofactor expansion to compute $\det A$

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

Sol $\det A = a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33}$

$= (-1)^{3+1} \cdot a_{31} \cdot \det A_{31} + (-1)^{3+2} \cdot a_{32} \cdot \det A_{32} + (-1)^{3+3} \cdot a_{33} \cdot \det A_{33}$

$= 0 \cdot \begin{vmatrix} 5 & 0 \\ 4 & -1 \end{vmatrix} - (-2) \begin{vmatrix} 1 & 0 \\ 2 & -1 \end{vmatrix} + 0 \cdot \begin{vmatrix} 1 & 5 \\ 2 & 4 \end{vmatrix}$

$= 0 + 2 \cdot (-1 - 0) + 0 = -2$ //

Ex:

$A = \begin{bmatrix} 3 & -7 & 8 & 9 & -6 \\ 0 & 2 & -5 & 7 & 3 \\ 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 2 & 4 & -1 \\ 0 & 0 & 0 & -2 & 0 \end{bmatrix}$ Calculate $\det A$.

$\det A = 3 \cdot \begin{vmatrix} 2 & -5 & 7 & 3 \\ 0 & 1 & 5 & 0 \\ 0 & 2 & 4 & -1 \\ 0 & 0 & -2 & 0 \end{vmatrix} - 0C_{21} + 0C_{31} - 0C_{41} + 0C_{51}$

$= 3 \cdot 2 \begin{vmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{vmatrix} - 0 \cdot C_{21} + 0 \cdot C_{31} - 0 \cdot C_{41}$

$= -2$

$= 3 \cdot 2 \cdot (-2) = -12.$

Theorem: If A is a triangular matrix, then $\det A$ is the product of the entries on the main diagonal of A.